Cavern Escape

A fast-paced 2D platformer where players jump, dodge hazards, and defeat enemies while navigating increasingly challenging levels to reach the exit and survive.

Version History

Date	Version	Changes
3/4/2026	0.1.0	Implemented Tilemaps and Rules
3/6/2026	0.1.1	Painted Scenes
3/10/2026	0.2.0	Player Movement & Physics, Player Animation, Camera
3/12/2026	0.2.1	Layers & Collision Matrix, Enemy AI
3/14/2026	0.2.2	Shooting System, Health & Damage
3/18/2026	0.3.0	Scenes & Game States
3/20/2026	1.0.0	Audio & Polish
3/26/2026	1.0.1	Added +1 on enemy, conditional level passing, enemy counter

Date: 03/26/2026

Project Info

Game Title: Cavern Escape
Author: Justin Fok
Platform: PC
Engine / Tools: Unity (2D), C#

Game Overview and Vision

High Concept:

A simple but engaging 2D platformer focused on movement, survival, and progression through increasingly dangerous levels.

Game Summary:

The player controls a character who must traverse levels filled with enemies and hazards. The goal is to reach the exit while avoiding damage and managing limited health. The game emphasizes responsive controls, simple mechanics, and clear progression.

Player Journey Snapshot:

Early Game:

The player learns basic movement and jumping. They encounter simple enemies and hazards and understand how to reach the level exit.

Late Game:

The player navigates more complex layouts with tighter jumps, more enemies, and hazards. Timing, positioning, and awareness become more important.

Unique Selling Points:

- Smooth and responsive movement controls
- Simple but satisfying combat/damage system
- Clean level progression system
- Lightweight but complete gameplay loop

Target Audience:

- Casual players
- Beginner platformer fans
- Students learning game development

Genre & Inspirations

- Genre: 2D Platformer
- Inspired by: Mario, Celeste, classic platformers

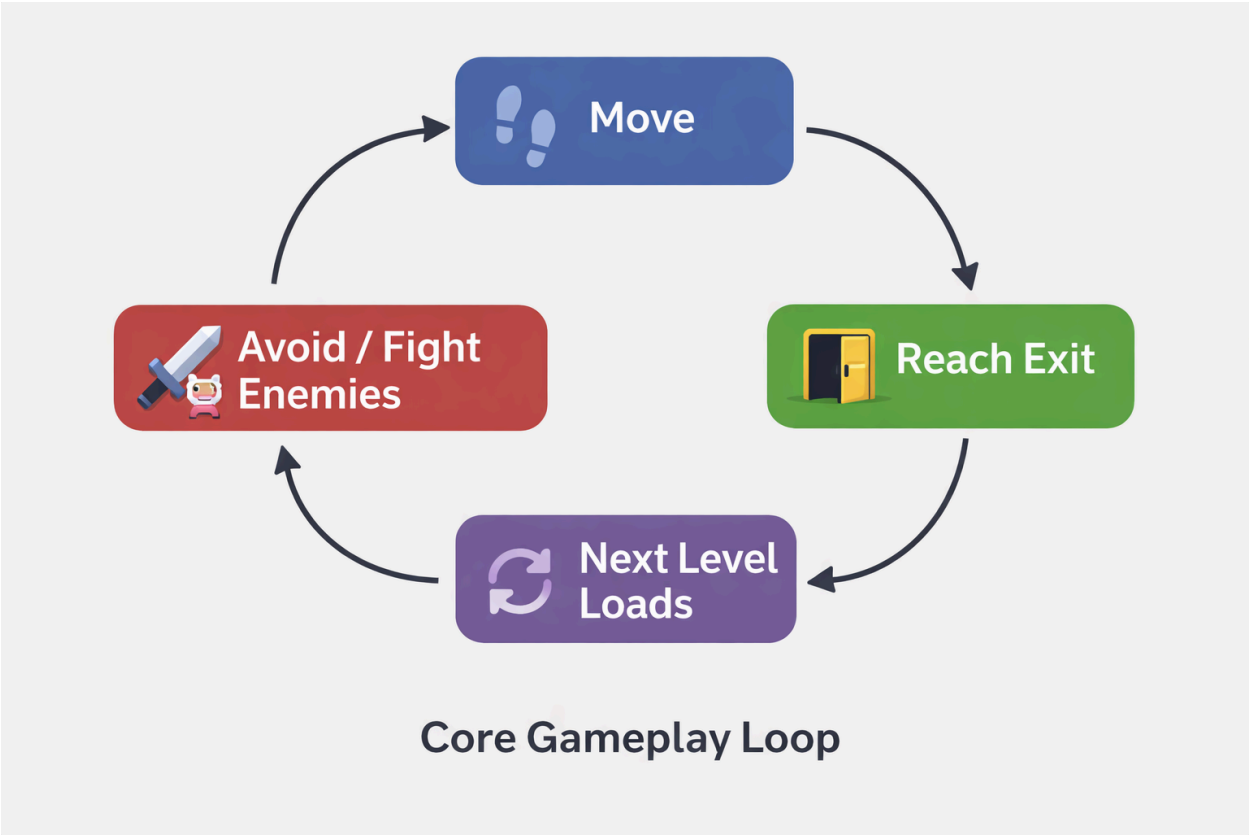
Tone & Look & Feel

- Pixel art style
- Bright, simple visuals
- Lighthearted but slightly challenging gameplay

Core Gameplay Design

Core Gameplay Loop

Move → Avoid/Interact → Reach Exit → Load Next Level → Repeat



Player Progression

- Progress through levels
- Increasing difficulty (more hazards/enemies)
- Health management becomes more important

Controls & Input

- A/D or Arrow Keys → Move
- Space → Jump
- Double Jump (optional feature added)

Difficulty & Challenge

- Starts easy
- Gradually increases with hazards and enemies
- Player skill-based (timing and movement)

Feedback Systems

- UI hearts for health
- Sound effects (jump, damage, etc.)
- Animations for movement and death
- Camera follow for smooth gameplay

Game Mechanics & Systems

Core Mechanics

- Movement (left/right)
- Jumping + double jump
- Enemy collision damage
- Hazard damage (instant kill)
- Level exit trigger

Side Mechanics

- Camera follow (Cinemachine)

Interaction Mechanics

- Collision-based gameplay
- Trigger zones (exit, hazards)
- Physics-based movement

Economy & Resources

- Health system (hearts)
- Lives system (GameSession)

Narrative, World, & Characters

Themes

- Survival
- Progression
- Challenge

Story Overview

Minimal narrative. The player progresses through levels overcoming obstacles and enemies to reach the end.

Worldbuilding

- Simple platforming environment
- Floating platforms and hazards

Main Characters

- Player character (controlled by user)
- Enemies (basic patrol units)

Player Agency
- Movement choices
- Jump timing
- Path navigation

Level & Encounter Design
Objectives: Reach the level exit

Structure & Layout
- Linear platforming levels
- Platforms at varying heights

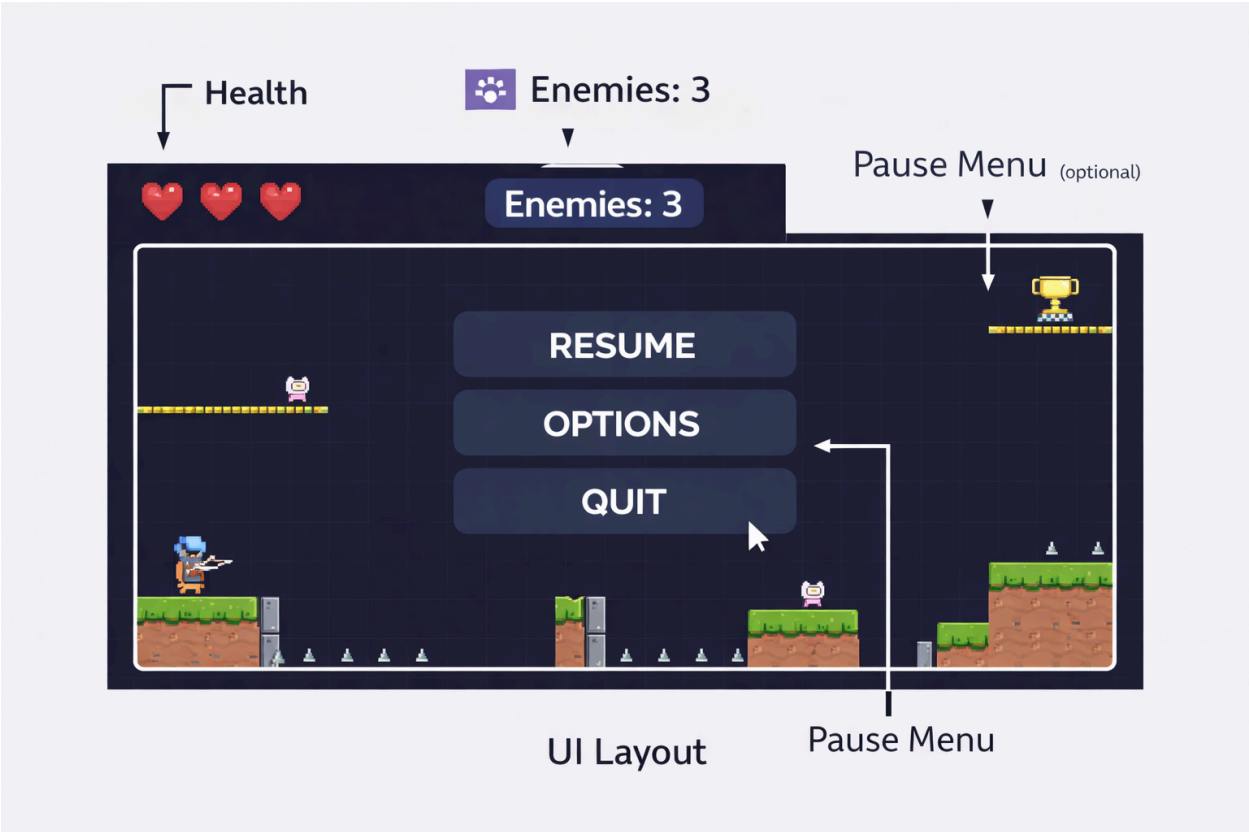
Gameplay Flow: Start → Jump between platforms → Avoid hazards → Reach exit
Interactive Elements
- Enemies
- Hazard tiles
- Exit object

Progression Gates: Level completion required to advance

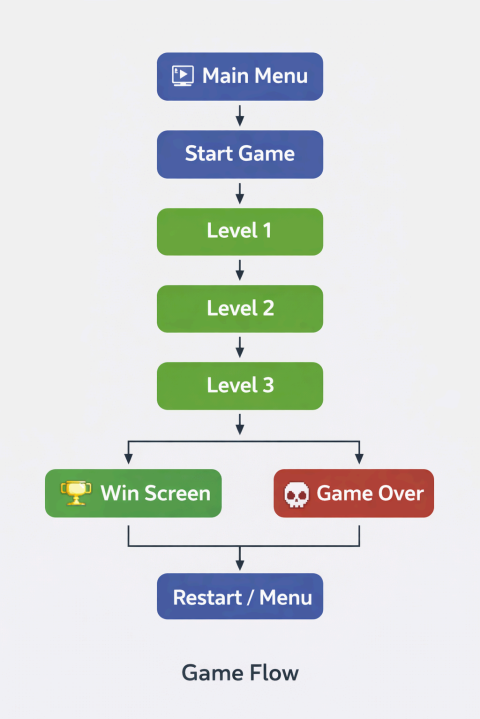
UI, UX, & Accessibility
Core UI Elements
- Health hearts
- Game over screen

Menus & Screens
- Main Menu
- Game Over
- Win Screen

HUD Design
- Hearts displayed on screen
- Minimal UI for clarity



UX Flow: Main Menu → Level 1 → Level 2 → Level 3 → Win / Game Over



Onboarding

- Learn by playing
- Simple mechanics introduced early

Accessibility Options

- Basic controls
- Simple visuals

Production Notes

Milestones

- Prototype complete
- Core gameplay implemented
- UI + Audio added

Out-of-Scope Features: Multiplayer

Future Improvements

- Better animations
- More levels
- Improved enemy behavior
- Checkpoints